team-arcane

Midterm Evaluation: Team Arcane (ISM6562ArcaneMidterm)

Team Members: Jonathan James, Nairi Keeney, Maaz Ul Hasan Masood, Abdul Mannan Mohammad **Repository:** https://github.com/nairikny/ISM6562ArcaneMidterm **Level:** Graduate (ISM 6562)

Part 1: Operational Databases, Data Warehousing, and ETL (20 points)

Section 1: Environment Setup (3 pts)

Task	Points	Score	Notes
1.1 Docker compose ps (5 services)	1	1	All five services documented
1.2 \dt for all three databases	1	1	Tables listed for sales, HR, warehouse
1.3 pgAdmin screenshot (3 servers)	1	1	Screenshot showing all three servers connected

Subtotal: 3/3

Section 2: Explore Operational Databases (4 pts)

Task	Points	Score	Notes
2.1 Sales schema \d + normalization response	1.5	1.5	Output shown; response correctly identifies PKs, FKs, and normalization to 3NF
2.2 HR schema \d + relationships response	1.5	1.5	Output shown; correctly explains one-to-many relationships and composite unique constraint
2.3 Four provided queries with output	1	1	All four query outputs present

Subtotal: 4/4

Section 3: Data Warehouse Concepts (4 pts)

Task	Points	Score	Notes
3.1 Star schema \d for 5 tables	0.5	0.5	All five warehouse table structures shown
3.2 Operational vs warehouse comparison (200-250 words)	2	2	Addresses all three areas: structural differences, denormalization rationale, dim_date purpose
3.3 Facts vs dimensions identification (150-200 words)	1.5	1.5	Correctly identifies fact_sales as fact table, others as dimensions; grain identified; good example business question

Subtotal: 4/4

Section 4: ETL Pipeline (5 pts)

Task	Points	Score	Notes
4.1 ETL script analysis (200-250 words)	2	2	Covers extract, transform, load, and idempotency; mentions cron schedule correctly
4.2 Count verification	1	1	Source and warehouse counts shown and match
4.3 New order through ETL	2	2	Insert, verify, trigger ETL, verify in warehouse -- all documented

Subtotal: 5/5

Section 5: Analytical Queries (4 pts)

Task	Points	Score	Notes
5.1 Query 1: Monthly sales summary	1	1	Correct SQL with proper JOINs and GROUP BY
5.1 Query 2: Top customers by spending	1	1	Correct SQL and output
5.1 Query 3: Revenue by category and month	1	1	Correct multi-table JOIN query
5.2 EXPLAIN ANALYZE comparison	1	1	Both outputs shown; written response discusses scalability

Subtotal: 4/4

Part 1 Total: 20/20

Part 2: Sharding the Sales Database (20 points)

Section 1: Setup & Exploration (4 pts)

Task	Points	Score	Notes
1.1 Docker compose ps (6 services)	1	1	All six services running

Task	Points	Score	Notes
1.2 Count rows in each shard	1	1	SE: 7 customers, 24 orders; NE: 1 customer, 3 orders
1.3 Uneven distribution analysis (150-200 words)	2	2	Identifies hot shard problem; suggests hash-based partitioning as alternative

Subtotal: 4/4

Section 2: Update ETL Pipeline (8 pts)

Task	Points	Score	Notes
2.1 Diagnose ETL failure (100-150 words)	2	2	Correctly identifies missing SALES_DB_HOST; describes dual-shard approach
2.2 Modified ETL script + output	4	4	Working script in repo; correctly connects to both shards, merges results; products from SE only
2.3 Verify warehouse data + new order	2	2	Counts verified (8 customers, 10 products, 27 orders); new order flows through

Subtotal: 8/8

Section 3: Cross-Shard Challenges (4 pts)

Task	Points	Score	Notes
3.1 Cross-shard query + limitation documented	2	2	Both query outputs shown; correctly documents inability to cross-shard JOIN
3.2 Sharding trade-off analysis (200-250 words)	2	2	Good coverage of benefits (locality, isolation, scaling) and drawbacks (no cross-shard queries, ETL complexity, hot shards)

Subtotal: 4/4

Section 4: EXPLAIN ANALYZE (4 pts)

Task	Points	Score	Notes
4.1 SE shard EXPLAIN ANALYZE	2	2	Output shown
4.2 Warehouse EXPLAIN ANALYZE + written response	2	2	Warehouse isolation from sharding well explained

Subtotal: 4/4

Part 2 Total: 20/20

Part 3: Adding Cassandra for Real-Time Order Ingestion (20 points)

Section 1: Setup & Cassandra Exploration (4 pts)

Task	Points	Score	Notes
1.1 Docker compose ps (8 services)	1	1	All eight services documented

Task	Points	Score	Notes
1.2 Nodetool status + DESCRIBE outputs	1	1	Two UN nodes; keyspace and table descriptions shown
1.3 Cassandra vs PostgreSQL table design (150-200 words)	2	2	Good explanation of denormalization and query-first design with two tables

Subtotal: 4/4

Section 2: Query-First Design Analysis (4 pts)

Task	Points	Score	Notes
2.1 Valid Cassandra queries (2 queries)	2	2	Both query outputs present
2.2 Invalid query error + explanation (100-150 words)	2	2	Error documented; partition key restriction well explained

Subtotal: 4/4

Section 3: Write Orders to Cassandra (4 pts)

Task	Points	Score	Notes
3.1 Dual inserts + verification	2	2	Both inserts and verification queries shown
3.2 Dual-write consistency discussion (100-150 words)	2	2	Identifies inconsistency risk; mentions logged batches

Subtotal: 4/4

Section 4: Update ETL Pipeline (4 pts)

Task	Points	Score	Notes
4.1 ETL analysis (100-150 words)	2	2	Correctly identifies needed changes
4.2 Modified ETL code + verification	2	2	Cassandra extraction function present in docx; warehouse verification shown

Subtotal: 4/4

Section 5: Polyglot Persistence Reflection (4 pts)

Task	Points	Score	Notes
5.1 Comparison table + response (200-250 words)	2	2	Table covers all aspects; CAP positioning correct (PG=CP, Cassandra=AP)
5.2 Cassandra as warehouse argument (200-250 words)	2	2	Argues against; correctly cites lack of JOINs, poor GROUP BY, eventual consistency

Subtotal: 4/4

Part 3 Total: 20/20

Part 4: Distributed SQL Database Presentation (40 points)

> **Note on scoring:** Full marks awarded for the in-class presentation delivery portion (36 points: Technical Accuracy, TechRetail Evaluation, Presentation Quality, Visual Aids, Q&A Handling, Equal Participation). While there was room for improvement on each presentation, the feedback given and discussions held during class provide overall guidance for areas of growth. Full marks reflect the lengthy delay between the presentations and these evaluations being returned.

Category	Points	Score	Notes
Technical Accuracy	8	8	Full marks.
TechRetail Evaluation	8	8	Full marks.
Presentation Quality	8	8	Full marks.
Visual Aids	6	6	Full marks.
Q&A Handling	2	2	Full marks.
Equal Participation	4	4	Full marks.
Slide Content (/10)	—	**0**	**No presentation file submitted to repo.** Cannot evaluate content.
GitHub Repository	**4**	**2**	Repo organized into part01/02/03; prof-tcsmith has access. **Two rubric violations:** (1) Only 3 of 4 team members have commits (Abdul Mannan: 0 commits — rubric requires every member to have at least one commit); (2) No Part 4 slide deck in repo. -1 per violation.

Part 4 Total: 38/40

GitHub Repository Assessment

Criterion	Status
Repo created and organized by parts	Yes (part01/, part02/, part03/)
Prof-tcsmith invited as collaborator	Yes (origin remote = prof-tcsmith fork)
All team members have >= 1 commit	**No** -- Abdul Mannan Mohammad has 0 commits
Part 4 slide deck in repo	**No** -- not found
Submission docs present	Yes -- .docx files for all 3 parts
Modified ETL scripts included	Yes -- Part 2 etl_script.py in repo

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Summary

Component	Points Available	Score
Part 1 -- Operational Databases & ETL	20	20
Part 2 -- Sharding	20	20
Part 3 -- Cassandra	20	20
Part 4 -- Repository	4	2
Part 4 -- Presentation delivery (in-class)	36	36
Total	**100**	**98/100**

Strengths

- Excellent, thorough work across all three hands-on parts
- ETL modification for Part 2 is clean and correct
- Written responses demonstrate genuine understanding of concepts
- Good documentation throughout

Areas for Improvement

- Abdul Mannan Mohammad needs to contribute at least one commit to the repo
- Part 4 slide deck should be committed to the repository
- Consider more detailed commit messages (many "Add files via upload")